

Evaluation of foundation model performance in chest radiology tasks based on pre-training data modality

Taejoon Park

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parkt@carleton.edu



Carleton



MASSACHUSETTS
GENERAL HOSPITAL



HARVARD
MEDICAL SCHOOL

Background

- Chest X-ray imaging
 - Remains an accessible modality across clinics of various resource levels.
 - Contains low-level features that indicate a wide variety of thoracic diseases.
 - Are processed in large volumes by radiologists.
- Deep learning methods seek to reduce load on radiologists as well as improve recall of unexpected diseases.¹
- Foundation models have emerged in clinical research as an advancement in various deep learning tasks.²
 - Employ large pre-training data.
 - Can be efficiently fine-tuned for downstream clinical tasks.

¹ Milam. *CR*. 2023.

² Zhang. *MIA*. 2024.

Central question

- Foundation models are distinguished by the pre-training data domain.³
- Do chest x-ray specific foundation models outperform general biomedical information foundation models?

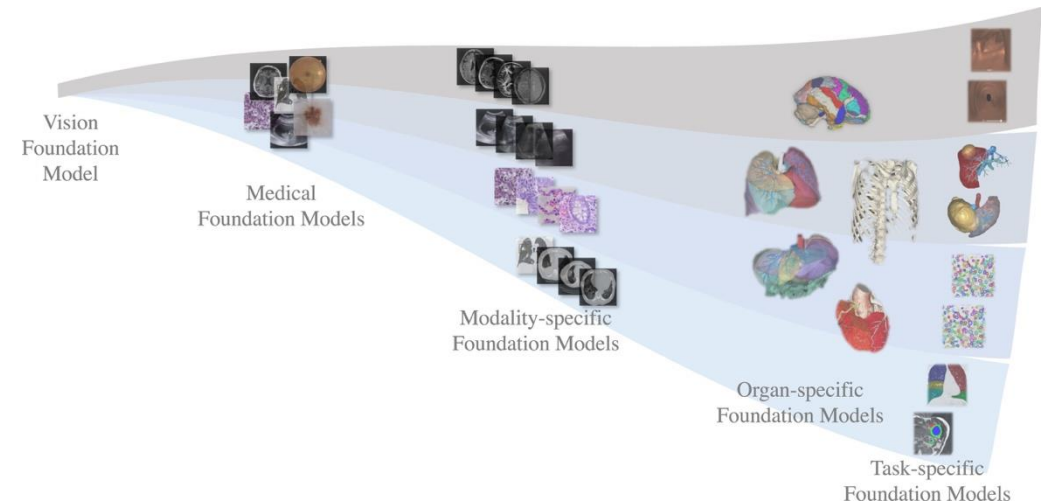


Fig. 1. The spectrum of foundation models in medical image analysis.⁴

Models

- MedSigLIP⁵
 - SigLIP architecture
 - Pre-trained on general biomedical text-image pairs: *chest X-rays, dermatology images, ophthalmology images, histopathology slides, and slices of CT and MRI volumes, along with associated descriptions or reports.*⁵
- CheXagent⁶
 - SigLIP architecture
- CXR Foundation⁷
 - *EfficientNet-L2 architecture and BERT architecture*
- BioViL-T⁸
 - Image: Vision Transformer and ResNet-50; Text: BERT
- CheXFound⁹
 - ViT-L architecture
- EVA-X¹⁰
 - ViT-L architecture

⁵ Google. *HAI-DEF*. 2025.

⁶ Chen. *arXiv*. 2025.

⁷ Google. *HAI-DEF*. 2026.

⁸ Bannur. *CVPR*. 2023.

⁹ Yang. *IEEE*. 2024.

¹⁰ Yao. *DM*. 2025.

Data source

- Images, image metadata, and structured disease labels obtained from MIMIC-CXR-JPG (v. 2.1.0)¹¹
 - JPG files from full DICOM information
 - 377,110 de-identified chest x-ray images from the Beth Israel Deaconess Medical Centre¹²
 - Accessed through PhysioNet¹³, required credentialing and data handling training
 - Contains structured labels derived from associated text reports using CheXpert (AIMI) and NegBio (NIH) natural language processing methods (1: positive, 0: negative, -1: uncertain, -2: unmentioned)
 - Fourteen categories from Atelectasis to No Finding
- Text reports obtained from MIMIC-CXR (v. 2.1.0)¹⁴
 - Parent of MIMIC-CXR-JPG
 - 227,835 structured text reports accompany radiology images (multiple images to one report/study)
 - Sourced from de-identified electronic health records at the Beth Israel Deaconess Medical Centre

¹¹ Johnson. *PhysioNet*. 2024.

¹² Johnson. *arXiv*. 2019.

¹³ Pollard. *NH*. 2026.

¹⁴ Johnson. *PhysioNet*. 2024.

Data pre-processing

- Quality control
 - Images ranked by several metrics:
 - Lowest half-image mean value
 - Lowest variance
 - Most extreme mean pixel value
 - Rankings used to hand-curate a mini dataset of 50/50 problematic and clean images n=50
 - Pre-trained *Ultralytics YOLO26 nano* model¹⁵ trained on classification on the mini-dataset
 - Final manual thresholding was applied to identify 85 problematic images
 - Soft delete applied
- Report text split into “Impression” and “Findings” sections
- Official test splits used for all evaluation
- Metadata used:
 - dicom_id
 - study_id and subject_id
 - ViewCodeSequence_CodeMeaning



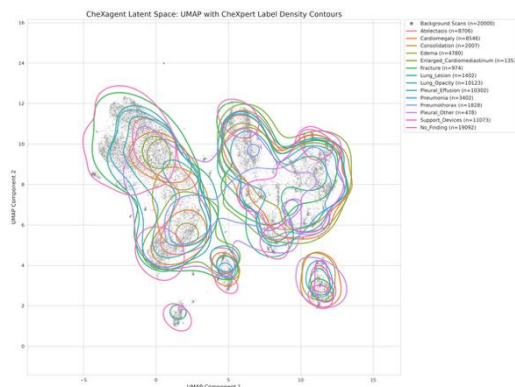
Foundation model inference

- Each foundation model was evaluated with Python within an isolated virtual environment
 - Each virtual environment was set up with dependencies as recommended by each
 - All inference was done on the tauon HPC
- For models with multiple variations, the best-performing model as reported for each task was selected
- For image-text models, image and text embeddings were created within the projected joint latent space
- Images were reshaped according to each model architecture
- Texts were tokenised with provided methods with respect to model token limits

Experiments

- **Structured label classification**

- Linear probing of image embeddings based on provided structured labels (linear regression using PyTorch)
- Understand the organisation and definition of image embedding space
- Metrics: AUROC, AUPRC
- Models: MedSigLIP, CheXagent, CXR Foundation, BioViL-T, CheXFound, EVA-X



- Sensitivity analyses

- extreme gradient boost classification, MLP classification (shared and independent architectures)
- L2 normalisation, principal component analysis

- Early stopping of 5 epochs without 0.0001 of validation improvement

- Hybrid fixed grid hyperparameter search for MLP classifier heads

- 1 or 2 hidden layers
 - k_1 set to $\alpha * \text{input_size}$ (based on embedding)
 - k_2 is constant compression ratio of k_1

- Limited to postero-anterior view

Experiments cont.

- **Zero shot classification**

- Disease category classification based on cosine similarity between disease phrase and image embeddings
- Understand the rigour of image-text joint latent space
- Metrics: AUROC, AUPRC
- Models: MedSigLIP, CXR Foundation, BioViL-T

- Positive-negative phrase pairs created for each disease category

- eg. “radiographic findings consistent with atelectasis”; “no evidence of atelectasis, lungs are fully inflated”

- **Image-report recall**

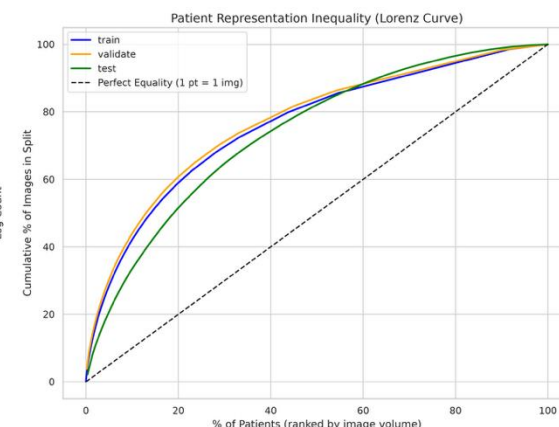
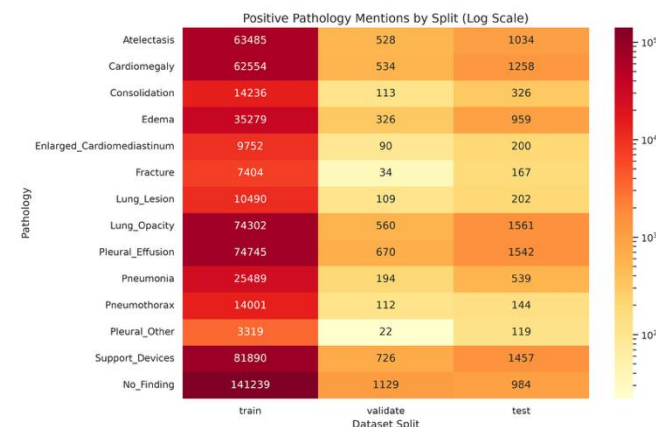
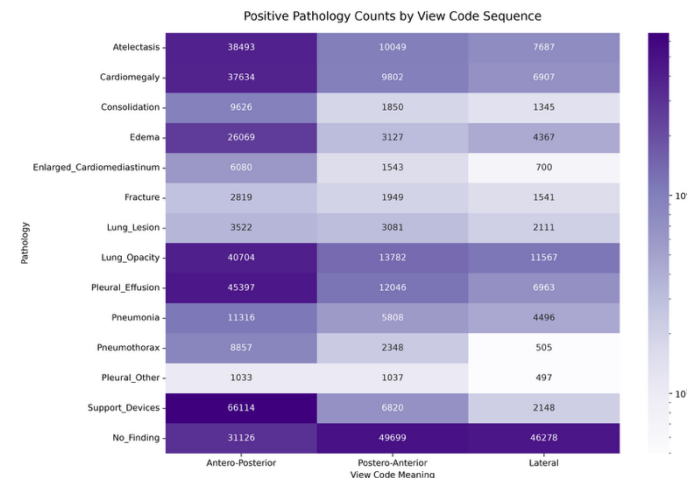
- Attempt to match corresponding images to report and vice versa by ranking cosine similarity
- Metrics: Recall@n, MRR
- Models: MedSigLIP, CXR Foundation, BioViL-T

- Variations:

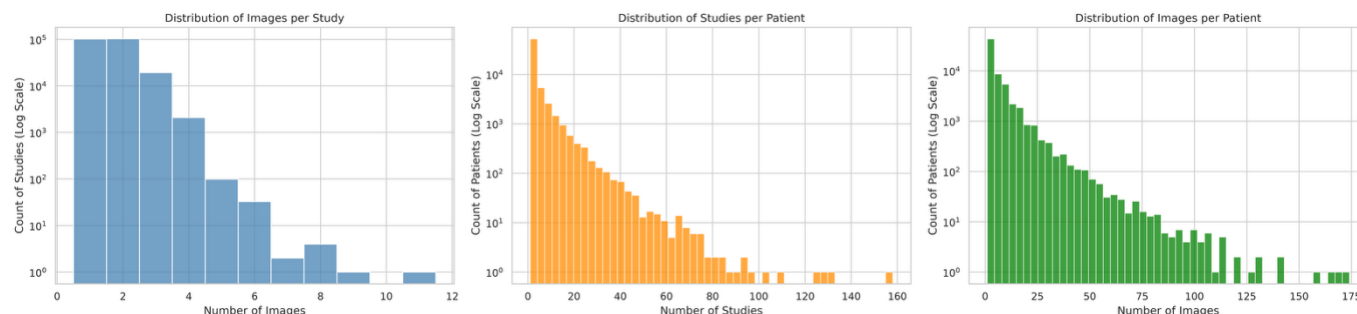
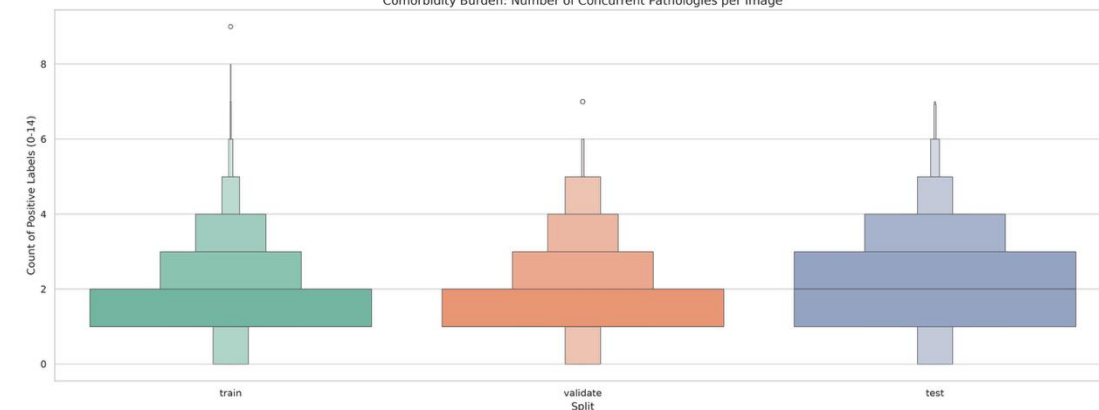
- Multiple image to one report
- Reference centroid of all images of a particular study as image embedding
- Reference centroid of impression and findings section embeddings as text embedding

Data demographics

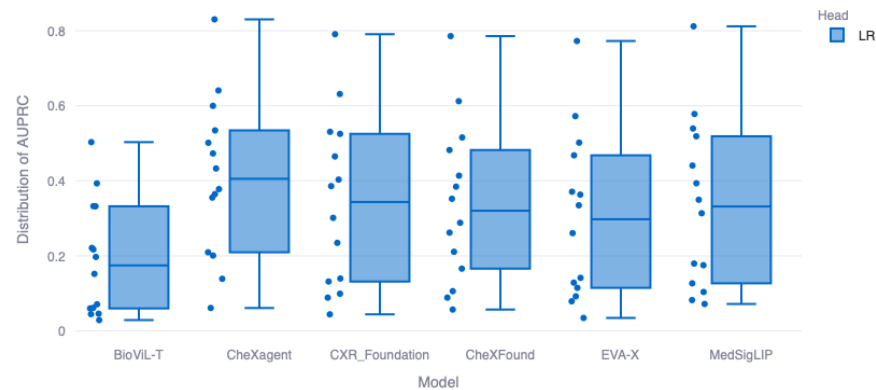
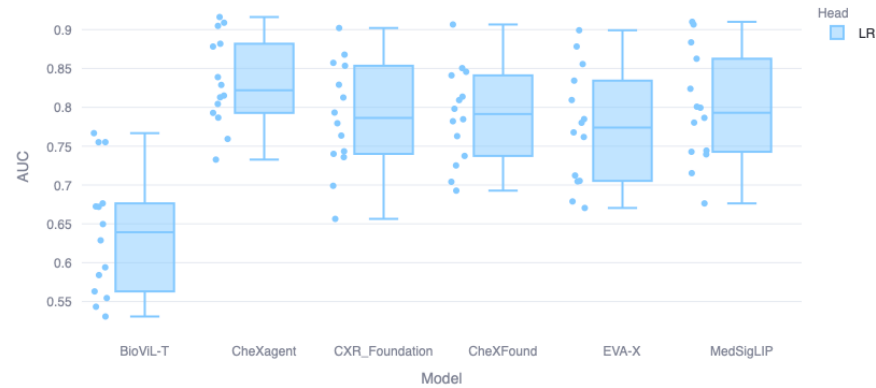
- Individual patients were not overrepresented in the data to a problematic degree
- Patients were cleanly split across train/val/test splits ensuring no data breach
- Comorbidity was not present to an extreme degree



Comorbidity Burden: Number of Concurrent Pathologies per Image

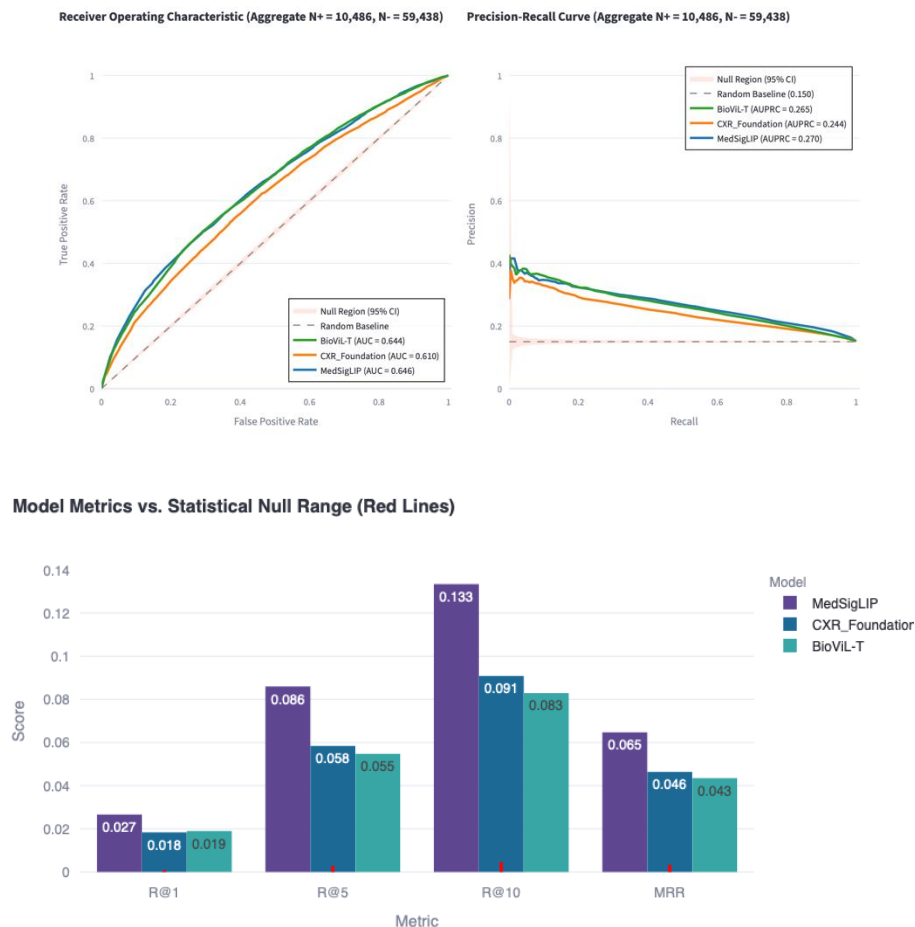


Primary outcomes



- MedSigLIP shows insignificantly divergent performance compared to CXR domain foundation models in linear probing AUROC and AUPRC
- CheXagent showed marginally superior performance in linear probing

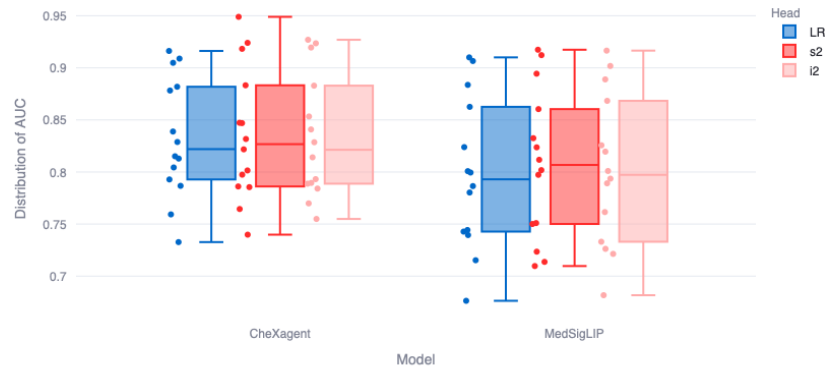
Primary outcomes (cont.)



- Zero shot classification performance is further comparable between general biomedical domain vs. CXR domain foundation models
- General biomedical domain model's superior performance is observed in image-to-report and report-to-image matching

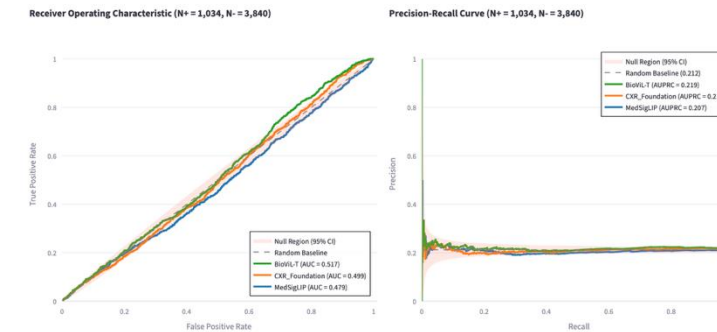
Secondary outcomes

- MLP classifiers did not show consistently higher performance than linear probing
- Certain disease categories such as atelectasis showed poor performance across all models and methods



Atelectasis Curves (Chexpert Labels)

Settings: Uncertainty (-1) = U-ignore (Drop -1) | Unmentioned (-2/NaN) = Map -2 / NaN to 0 (Presume Absent)



Discussion

- MedSigLIP shows insignificant divergence in linear probing AUROC/AUPRC compared to CXR specific models
 - Data diversity in pre-training is a critical factor alongside data scale¹⁶
 - Pre-training across diverse imaging modalities allows models to learn fundamental visual primitives that transfer to thoracic radiology
 - CheXagent (SigLIP) achieved marginally superior results, demonstrating domain-specific alignment is a slight advantage
- Non-linear MLP and XGB classifier heads failed to consistently outperform simple linear regression
 - Confirms foundation model pre-training already yields cleanly organised latent feature spaces

Discussion cont.

- MedSigLIP significantly outperformed CXR-specific models in image-report retrieval
 - CXR-only language models may overfit to formulaic and templated radiology reports
 - Pre-training on broader medical literature may create a richer and more flexible semantic space
- All evaluated models shared similarly weak performance on certain disease labels
 - Atelectasis: well-represented, large features
 - Fracture: slightly under-represented, pixel-level features
 - Potentially a shared architecturally weakness in down-sampling of images, losing boundaries, and patch-wise tokenisation, losing large features

Future directions

- Dense latent probing
 - Evaluate token-level dense feature maps rather than globally pooled embeddings
 - Improved clinical explainability through phrase grounding
 - Potentially improve spatial resolution for preserving small features
- Targeted fine-tuning via LoRA
 - Efficient fine-tuning on curated datasets to improve performance on difficult and underperforming disease labels (eg. fracture)
 - Prevent catastrophic forgetting by freezing base weights
- Hybrid pre-training with data curation
 - First stage, general biomedical information pre-training
 - Second stage, implement a domain-specific pre-training phase using prototype-driven data selection techniques¹⁷
 - Maximising data diversity during domain-specific pre-training prevents latent space collapse and over-indexing on repetitive boilerplate reports, maintaining the broad multi-modal capabilities learned during initial generalist pre-training.

¹⁷Wang. *arXiv*. 2026.

Future directions cont.

- Report generation
 - Do CXR-specific foundation models outperform general biomedical information foundation models in structured report generation?
- External validation
 - Do generalist models demonstrate superior classification (zero-shot and linear probing) generalisation when exposed to out of distribution institutional data?

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Data availability

- All models and data used in these experiments are publicly available online at respective locations.
- All code for these experiments can be found at https://github.com/tpark747949/cxr_foundation_eval with documentation coming soon.